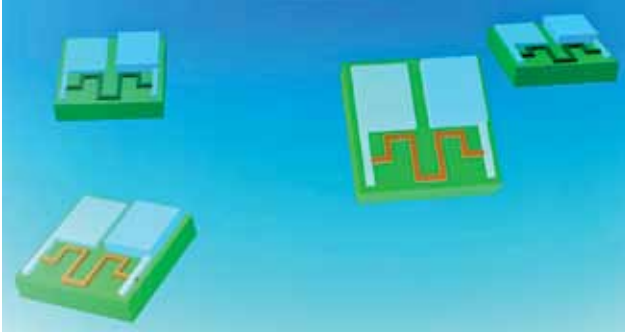


Human-hair thick battery for autonomous micro-robots

MIT engineers have developed a tiny battery, just 0.1mm long and 0.002mm thick—comparable to a human hair—that could revolutionise autonomous, cell-sized robots for medical uses like drug delivery and detecting gas leaks.



The zinc-air battery is 0.1mm long and 0.002mm thick (Credit: Courtesy of the researchers)

This zinc-air battery captures oxygen from the air to generate a 1-volt current, enough to power tiny circuits, sensors, or actuators. The team is integrating robotic functions into the battery itself, enabling micro-robots to operate independently in inaccessible spaces, unlike previous solar-dependent designs. This innovation could transform miniature robotics for various applications.

RF-VLC tech for line-of-sight communication

Researchers from Central University (CU), IIDM, and CU J&K in India have developed a hybrid RF-VLC communication system aimed at enhancing indoor communication by offering high data transmission rates while simultaneously reducing energy consumption. This system integrates radio frequency and visible light communication (VLC) to provide both illumination and communication, optimising power usage according to user demand while maintaining quality of service (QoS) and quality of experience (QoE) for various applications. The system comprises a transmitter, which sends binary data through light produced by LEDs, and a receiver, equipped with a photo-sensitive device like a photodiode or camera, that captures the data from the light when in direct line of sight to the transmitter. The team has also introduced modulation schemes that ensure consistent data flow and power use.

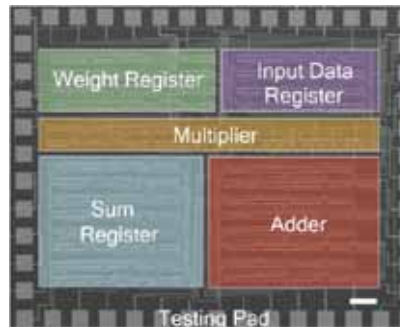
Energy-efficient 2D FET

Engineers from the Shanghai Institute of Microsystem and Information Technology, City University of Hong Kong,

and Fudan University have developed a new low-power, two-dimensional field-effect transistor (FET). The research addresses key challenges like high gate leakage and low dielectric strength, which have hindered silicon-based FET miniaturisation. Using ultrathin single-crystalline aluminium oxide as a dielectric, the team created an FET with high-quality interfaces. This advancement could lead to smaller, more efficient chips, crucial for 5G, AI, and Internet of Things (IoT) applications.

First carbon nanotubes tensor chip

Researchers at Peking University have developed a carbon nanotube (CNT)-based tensor processing unit (TPU), marking a breakthrough in energy-efficient chip design. This



Schematic diagram illustrates the structure of a carbon-nanotube-based tensor processing unit (Courtesy of Peking University)

TPU addresses the limitations of silicon-based processors in AI applications by leveraging CNTs' unique properties. Built on a systolic array architecture with 3000 CNT field-effect transistors, the chip efficiently handles matrix multiplications while minimising energy consumption. It powers a convolutional neural network with 88% accuracy in image recognition, consuming just 295μW. Operating at 850MHz and achieving energy efficiency over 1 TOPS/W, this TPU could revolutionise AI processing with improved performance and reduced power demands.

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Ultra-stable gradient NMC cathode batteries

Researchers are advancing battery technology by enhancing cathode materials, focusing on layered oxide cathodes to improve performance and sustainability. Traditional cathodes degrade quickly at high voltages, but Argonne National Laboratory's team has developed ultra-stable NMC (nickel-manganese-cobalt) cathodes. These use a full-gradient design, where nickel concentration decreases and manganese increases from core to surface, improving stability and energy density. Their design, patented in 2012, also reduces cobalt content, lowering costs and environmental impact. These innovations allow batteries to operate at higher voltages (up to 4.7V), offering better performance, especially for electric vehicles and other next-gen applications.